



FOCUS**ON**FORCE

Deploying

Given a scenario, describe approaches to manage and deploy technical reference data.

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Table of Contents

-  [Technical Reference Data](#)
-  [Business Scenario](#)
-  [Requirements & Solutions](#)



After studying this topic, you should be able to:

-  Describe the different ways of storing and managing technical reference data in Salesforce.
-  Identify the considerations related to the use of technical reference data.
-  Describe approaches to manage and deploy technical reference data for various technical requirements.



Introduction

In Salesforce, **technical reference data** is used to categorize other data, standardize terminology, and ensure consistency across systems. It serves as a **reference** within applications, such as picklist values, configuration settings, or other standardized data sets.

Custom Metadata Types are typically used to store and manage technical reference data that can be deployed across Salesforce environments.

Various considerations are applicable to the management and deployment of technical reference data. For example, when using Salesforce objects to store the data and use it in Apex, **object relationships** and **governor limits** should be considered.



Technical Reference Data

Using Technical Reference Data

Technical reference data serves as a reference within applications, such as picklist values, configuration settings, or other standardized data sets. It is used for application functionality and not user-generated content.

Custom Metadata Types

A custom metadata type and its records can be created to store and manage technical reference data such as configuration settings for an application. Metadata API or Change Sets can be used for deployment.

Considerations

Important considerations for the management and deployment of technical reference data are related to the use of a source of truth, governance, data loading, object relationships, and governor limits.



Technical Reference Data

[Table of Contents](#)



Technical Reference Data

Technical reference data refers to data used across the organization to **categorize other data**, **standardize terminology**, and **ensure consistency** across systems.

❖ REFERENCE

Technical reference data serves as a **reference within applications**, such as **picklist values**, **configuration settings**, or other standardized data sets.

❖ USAGE

Technical reference data are crucial for **application functionality** but **not user-generated content**.

❖ DATA MANAGEMENT

A **single source of truth** can be utilized to manage technical reference data to ensure consistency and accuracy. For example, **Salesforce** can store the data, while a **version control system** can store the metadata.



Technical Reference Data

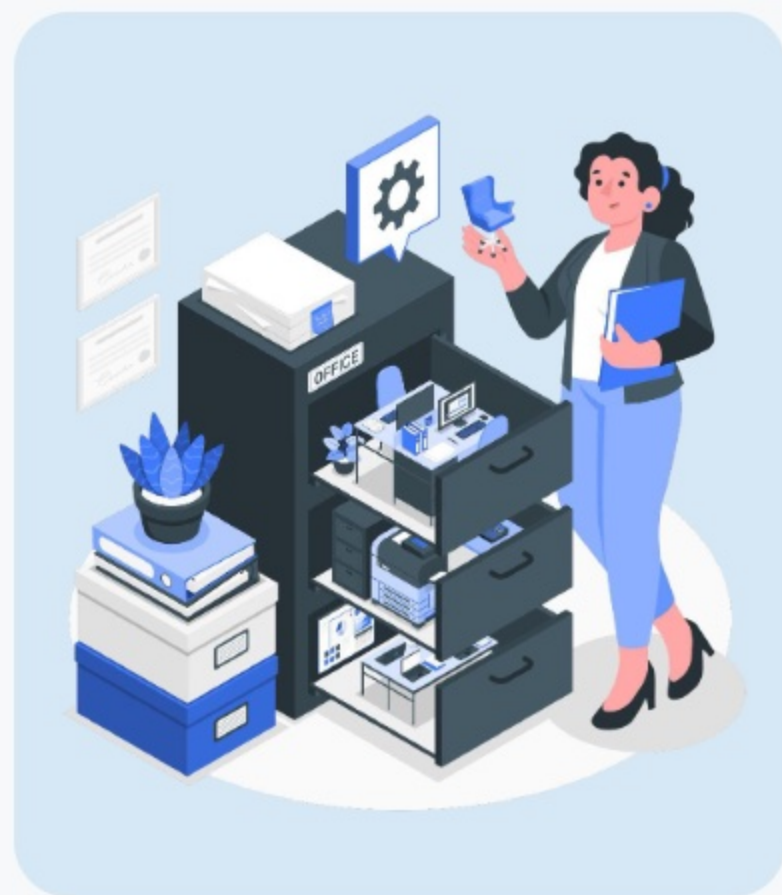
Technical reference data refers to data used across the organization to **categorize other data**, **standardize terminology**, and **ensure consistency** across systems.

❖ CUSTOM METADATA TYPES

Custom Metadata Types can be leveraged to **store and manage technical reference data** that can be deployed across environments.

❖ PICKLIST VALUES

Picklist values are used to **ensure data consistency** and **integrity** by **restricting field values** to a set of predefined options. Picklist values can be added or edited in a custom picklist field. A picklist field can also use values from a **global picklist value set**, which can be edited.



Technical Reference Data

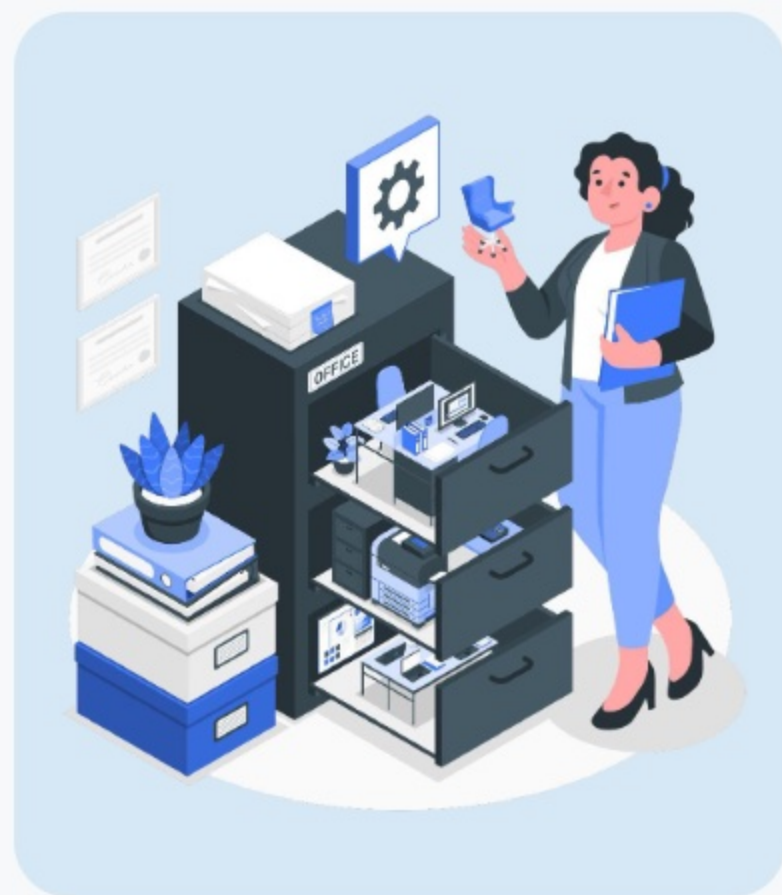
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❖ OBJECTS

Salesforce objects are often used to store and manage technical reference data. However, **object relationships and limitations**, such as **governor limits**, should be considered when using them in Apex.

❖ DEPLOYMENT

Metadata API can be utilized for the deployment of complex or large volume of technical reference data like custom metadata types and records. **Change Sets** can be utilized for simple deployments.



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❖ DATA LOADER

Large volumes of reference data in a Salesforce object can be **imported or exported** using **Data Loader**.

❖ GOVERNANCE

Strict governance policies and **version control** can be implemented to manage changes to reference data effectively.

❖ TESTING

Thorough testing and **validation** of reference data should be performed before deployment to production.



Business Scenario

Table of Contents



Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Health Innovations, a provider of digital health solutions, is transitioning to an enhanced operational model in Salesforce to consolidate patient management, healthcare analytics, and customer engagement across its diverse service platforms. As the company expands its reach in the global healthcare market and undertakes more intricate projects with the need for regular updates, it faces the necessity to refine its deployment methodology. The goal is to achieve streamlined, flawless deployments that are in harmony with the company's strategic objectives and the stringent compliance standards of the healthcare industry.

The company's Salesforce architect is tasked with evaluating the current needs for managing and deploying technical reference data for various use cases. The architect's role involves identifying optimal approaches to manage and deploy technical reference data critical for the company's operations.

Requirements & Solutions

[Table of Contents](#) 

Requirement 1



SCENARIO

The company is looking to implement a large number of features using an AppExchange product. The product allows the use of multiple related custom objects to store and configure important business logic within a custom application used for healthcare solutions. An administrator can create records of the custom objects to define the necessary business logic. The architect needs to recommend a suitable source of truth for managing this reference data and provide any relevant considerations related to loading a large volume of this data and using the data in Apex.



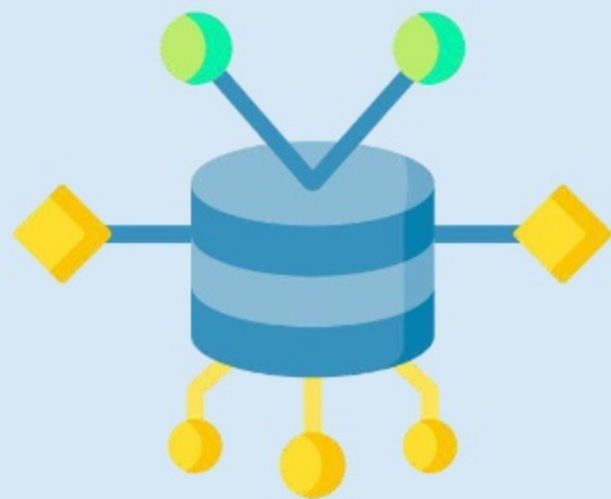
Solution 1 (Part 1)



SOLUTION

Salesforce can be used as the source of truth for the technical reference data stored in custom objects. The data can be stored in **production and sandbox environments**. For the metadata that defines the structure of these custom objects, a **version control system** such as Git can be used as the source of truth. **Data Loader** can be used to load a large volume of these records.

However, it is important to note that **circular relationships** become a concern in a data model that utilizes multiple related objects. It can lead to challenges in data integrity, updates, and navigation between related technical reference records. The records could indirectly reference each other in a loop, complicating updates and queries.

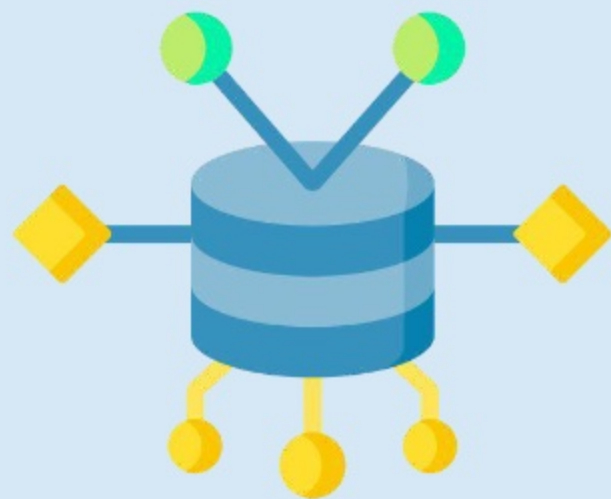


Solution 1 (Part 2)



SOLUTION

Another important consideration is that Salesforce imposes governor limits on the **depth of nested relationships** that can be accessed in a single SOQL query in Apex. When managing technical reference data across multiple objects, it's important to ensure that Apex transactions do not exceed these limits. Excessive nesting can hinder the ability to effectively utilize the reference data, impacting user experience.



Requirement 2



SCENARIO

The company's service department recently started using Salesforce. It would like to store and manage service centers, which are independent business units where customer service operations are performed. They are only required for reference and should be searchable in various web forms on the company's website and a custom application used for service management. The architect needs to recommend approaches for storing the service centers as technical reference data and their deployment.

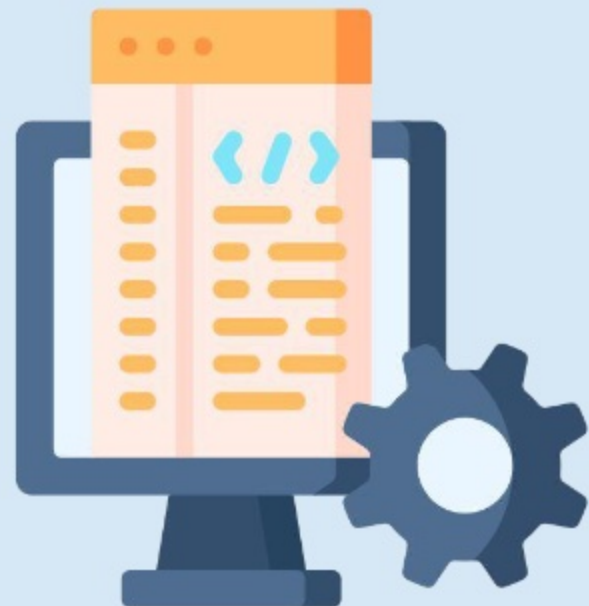


Solution 2



SOLUTION

A **custom metadata type** can be defined in Salesforce for the service centers. **Records** of this custom metadata type can be created to store and manage various service teams that should be searchable on web forms and the custom application used for service management. **Metadata API** or **Change Sets** can be utilized to deploy the custom metadata type and its records.



Learn More

-  [What are Custom Metadata Types?](#)
-  [Custom Metadata Types Basics](#)
-  [Execution Governors and Limits](#)

